Chapter 5, Problem 2P

Problem

The open-loop gain of an op amp is 50,000. Calculate the output voltage when there are inputs of +10 μV on the inverting terminal and +20 μV on the noninverting terminal.

Step-by-step solution

Step 1 of 2

Write the expression for output voltage of op-amp.

$$\begin{aligned} v_o &= Av_d \\ &= A(v_2 - v_1) \end{aligned}$$

Here,

A is the open loop voltage gain

v_1 is the inverting terminal voltage

v_2 is the inverting terminal voltage

Comments (1)

Anonymous

v2 non inverting terminal

Step 2 of 2

Calculate the output voltage.

$$v_o = A(v_2 - v_1)$$

Substitute 50,000 for A, $10\ \mu\text{V}$ for v_1, and $20\ \mu\text{V}$ for v_2 in the above equation.

$$\begin{aligned} v_o &= (50000)(20\times10^{-6} - 10\times10^{-6}) \\ &= (50000)(10\times10^{-6}) \\ &= 0.5\ \text{V} \end{aligned}$$

Therefore, the output voltage, v_o is $\boxed{0.5\ \text{V}}$.